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DiS MPHIL

Mechanistic Interpretability of Open Source LLMs

Reproduction of On the Biology of a Large Language Model

A REPORT PRESENTED

BY

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Abstract

The high accuracy of a linear probe is usually read as evidence that a concept is linearly decodable from the hidden states [1], but when the number of examples is far below the hidden dimension a linear classifier can separate the classes under any labelling, so the accuracy may reflect the geometry of a small sample rather than the representation [2]. We show, further, that the decoded direction is not always the one the model uses to compute its answer.

In a controlled binary task, the type of nuclear decay, on the Qwen3-4B model with per-layer transcoders, we identify at each layer three directions usually conflated into one, the *reading* direction w_{LDA} (the Fisher discriminant), the *writing* direction $\Delta\mu$ (the difference of class means), and the *use* direction u (the class-averaged gradient of the logit difference). The concept is decoded by w_{LDA} almost perfectly (AUC 0.98 to 0.99), yet intervention along it does not change the answer, while intervention along u or $\Delta\mu$ does. This inertness is controlled by geometry, the reading direction being causal only to the extent that it is non-orthogonal to the use direction, a degree that varies by concept, and with depth the use direction aligns with the unembedding contrast γ while the reading direction stays orthogonal to the output.

Relating the geometry to the transcoder features, the dictionary features write mainly the writing direction rather than the reading axis. At the cluster level they detect physical descriptors of the decay, the charge, the lepton number, the nuclear composition, and the type of transition, activating comparably on both classes, so they encode the physics of the process rather than the label, and the class is assembled from them late. The same separation of decoded and used appears at the feature level, the active dictionary spanning the writing direction well above a random null and the use direction partially, but the optimal reading axis no more than the null.

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1 Introduction

Linear representations and linear probing

The linear representation hypothesis states that interpretable features are encoded in the hidden states of neural networks as directions in the activation space [3, 4]. Building on this hypothesis, the standard interpretability method tests for the presence of a concept with a linear classifier, a probe, and high accuracy is interpreted as evidence that the concept is linearly represented [1, 5]. Probe methods have become an established tool, employed to detect part-of-speech, syntactic, and semantic structure [6, 7], as well as factual associations [8]. Their advantage is that the trained direction serves both as a test of the concept and as a candidate direction in which to intervene, and it is precisely this identification of the test with the mechanism that we question. There is no necessary connection between the *decodability* of a feature and its *causal role* in the computation, since a probe can reach high accuracy on a direction that correlates with the label yet is not used by the model when forming the output, measuring statistical separability rather than a computational mechanism [2, 9]. In response, causal methods have been proposed that assess the effect of a direction through intervention rather than through decoding [10, 11, 12]. The present work develops this distinction by showing, in a controlled task, that the direction from which the concept is decoded, the reading direction, and the direction to which the output is sensitive, the use direction, are not only different but approximately orthogonal, and that both this orthogonality and the causal inertness of reading depend on the concept.

Activation steering and the causal assessment of directions

An alternative to decoding is a causal assessment of a direction through steering. The chosen direction is added to the residual stream at a given layer, and the change in the model's behaviour is measured [13, 14, 15]. If the addition systematically shifts the output in the expected way, the direction is considered causally involved. A widely used construction of a steering direction is the difference of the mean activations between classes [16, 17]. This direction, which we denote $\Delta\mu$ and call the writing direction, is the between-class difference written into the residual stream. The third direction of interest is the class-averaged gradient of the margin, the logit difference between the class tokens, with respect to the residual stream. It identifies the direction in which a local change of the residual stream most shifts the model's decision [18]. This direction, which we denote u, is the direction to which the model's decision is locally most sensitive, and we take it as the use direction. Because the logits are a linear readout of the residual stream through the unembedding [19], the gradient of the logit difference lies along the unembedding contrast, which makes u a natural candidate for the direction through which the concept reaches the output.

Origin of the study

The project began as a reproduction of Anthropic's circuit tracing methodology [20, 21] on an open model, in which attribution graphs over per-layer transcoder features were constructed and validated by interventions (Appendix F). Two observations from that phase motivated the study of the residual stream directly, at the level of directions. Causal effects were carried by ensembles of features rather than by any single necessary pathway, and many of the features involved activated comparably on both classes, so neither the causal structure nor the class content could be read from individual features.

Transcoder features and hidden organisation

To connect the observed geometry with the model's computations, we use per-layer transcoders as an attribution method, sparse dictionaries that approximate the output of each layer's MLP block as a sum of interpretable feature directions [22, 20]. Whereas sparse autoencoders decompose a layer's activations into monosemantic features [23, 24], transcoders approximate the MLP sublayer

itself with a wider, sparsely activating layer, which allows its output to be attributed to individual features [25]. Since each feature contributes to the residual stream through its decoder direction, the features link the geometry of the directions to the content of the computation. They allow us to establish not only that the writing direction lies in the span of the transcoder features well above a random null, but also what content is written into it and how it relates to the use direction. Given the polysemanticity of features [26], a single feature is not a reliable unit of analysis, so we group features into clusters by their shared causal role and take the cluster as the working unit [21].

A controlled task

The central task is the binary classification of nuclear decay type. Given a description of the process, the model answers α or β. The same concept is presented in three surface forms of increasing context, a direct question, the same question with an answer cue, and the same question with two worked examples and an answer cue. Comparing them separates what is invariant to the surface form from what depends on it. Cross-concept generality is tested on an elementary particle classification, a multiclass task analysed as binary pairs. A full description of the datasets is given in Section 2.3.

Structure

Section 2 describes the methods, covering the model and transcoders, the four geometric objects, the feature metrics and cluster formation, and the intervention protocol and its metrics. **Section 3** presents the main results. These are the causal inertness of reading and its dependence on the concept, the formation of the writing direction by the MLP blocks, the qualitative dissociation of writing and use, the divergence of the directions across depth, the decision zone in which the separation arises, and the physical content of the writing direction. **Section 4** discusses the results in the context of interpretability and linear representations. The appendices contain the auxiliary experiments and additional results.

2 Methods

2.1 The model and the residual stream

All experiments are conducted on the pre-trained Qwen3-4B base language model [27]. The model is a decoder-only transformer [28] with $L = 36$ layers and hidden dimension $d = 2560$. We use the standard residual stream formulation of the transformer [19], in which the residual stream at position t is updated additively by the contributions of the attention and MLP blocks,

$$h_t^{(\ell)} = h_t^{(\ell-1)} + a_t^{(\ell)} + m_t^{(\ell)}, \tag{1}$$

where $h_t^{(\ell)} \in \mathbb{R}^d$ is the residual stream at the output of layer ℓ for position t, $a_t^{(\ell)}$ is the contribution of the attention block of layer ℓ, and $m_t^{(\ell)}$ is the contribution of its MLP block. The additive structure of (1) is essential to the analysis that follows, since it permits an exact decomposition of the change of any direction between adjacent layers into an attention contribution and an MLP contribution (Section 2.6). Throughout, we omit the position index where it is unambiguous, writing $h^{(\ell)}$.

The model's output over the vocabulary, the logits, is formed by applying the unembedding matrix $W_U \in \mathbb{R}^{|V|\times d}$ to the final normalised state. For a binary problem with classes c_0 and c_1, for example α and β, the quantity of interest is the logit difference between the class tokens.

2.2 Transcoders

To attribute the computation to individual features, we use per-layer transcoders [22, 20]. The layer-ℓ transcoder approximates the mapping from the MLP input to the MLP output of that layer as a sparse combination of feature directions. Given the MLP block input x, the transcoder computes feature activations

$$f^{(\ell)}(x) = \sigma\left(W_{\text{enc}}^{(\ell)} x + b_{\text{enc}}^{(\ell)}\right), \tag{2}$$

and reconstructs the MLP output as

$$\widehat{m}^{(\ell)} = W_{\text{dec}}^{(\ell)} f^{(\ell)}(x) + b_{\text{dec}}^{(\ell)}, \tag{3}$$

where $W_{\text{enc}}^{(\ell)} \in \mathbb{R}^{F\times d}$ and $W_{\text{dec}}^{(\ell)} \in \mathbb{R}^{d\times F}$ are the encoder and decoder matrices, F is the dictionary size, and σ is the activation nonlinearity. We use the pre-trained per-layer transcoders [29] with a dictionary of $F = 163{,}840$ features and a JumpReLU activation [30]. JumpReLU enforces threshold sparsity, a feature being active only if its pre-activation exceeds a trainable threshold, which yields a sparse set of active features for each input.

Each column $W_{\text{dec}}^{(\ell)}[:, j]$ defines the decoder direction of feature j, the direction along which feature j writes its contribution to the residual stream, and the activation $f_j^{(\ell)}$ sets the magnitude of this contribution.

Because the transcoders reconstruct the small class difference at low fidelity, with a relative error of 0.73 to 0.96, the analysis of the writing direction (Section 3.2) uses the true MLP output $m^{(\ell)}$ from (1) rather than its reconstruction $\widehat{m}^{(\ell)}$, so it does not depend on the reconstruction fidelity.

2.3 Datasets and prompt forms

The main corpus, which we call the α/β decay corpus, consists of descriptions of nuclear processes labelled with the decay type, α or β. It is class-balanced, with 269 examples of class α and 269 of class β, for a total of 538. Class balance gives the two class means equal estimation precision and keeps the pooled within-class covariance (Section 2.5.2) from being dominated by either class.

The same concept is presented in three prompt forms.

1. The **raw** form is a direct question about the decay type, with the class label as the target answer.

2. The **raw with suffix** form is a direct question with an added answer cue, a suffix that fixes the response format by requesting the answer as α or β.

3. The **scaffold** form prepends two labelled examples, each with the answer cue, before the target question [31]. These in-context examples fix the output format, indicating the expected answer form without otherwise changing the task (Section 3.2).

Because all three express the same α/β concept, agreement of the measured geometry across forms reflects invariance to the surface form rather than a coincidence of distinct concepts.

A second task tests cross-concept generality. The **elementary particle** task is a multiclass problem over electron, neutron, photon, and proton, with a corpus of 320 examples balanced across the four particles. From this corpus we form the six binary pairs, all pairs of the four classes, each balanced with 80 examples per class. The binary pairs test the invariance of the writing mechanism to the choice of pair within a single concept (Section 3.2).

2.4 Measurement position

All geometric objects are measured in the residual stream at the answer position, the position at which the model generates the class token. The prompt forms (Section 2.3) fix the output format so that this position is unambiguous and comparable across examples.

2.5 Four geometric objects

The analysis concerns four directions in the residual stream. Three are defined at each layer ℓ and differ in construction, mechanism, and causal role. The fourth, the unembedding direction γ, is layer-independent and serves as a reference for interpretation.

2.5.1 Writing direction $\Delta\mu$

The writing direction is defined as the difference of the mean residual stream between classes at layer ℓ,

$$\Delta\mu^{(\ell)} = \mu_{c_1}^{(\ell)} - \mu_{c_0}^{(\ell)}, \qquad \mu_c^{(\ell)} = \frac{1}{|D_c|} \sum_{x \in D_c} h^{(\ell)}(x), \tag{4}$$

where D_c is the set of examples of class c and $\mu_c^{(\ell)}$ is the class mean of the residual stream. Thus $\Delta\mu^{(\ell)}$ is the between-class difference written into the stream, the direction along which the mean representations of the two classes diverge [16, 17].

The difference of means construction subtracts the part of the representation common to both classes. The remaining direction $\Delta\mu$ still carries a substantial component along the mean residual stream $\bar{h}$, with $\cos(\Delta\mu, \bar{h}) \approx 0.59$ in the decision zone, so it is large in norm and partly aligned with the stream. For this reason we compare directions only after unit normalisation and track the norm of $\Delta\mu$ separately (Section 3.4). The normalised writing direction is $\widehat{\Delta\mu}^{(\ell)} = \Delta\mu^{(\ell)}/\|\Delta\mu^{(\ell)}\|$.

2.5.2 Reading direction $w_{\mathbf{LDA}}$

The reading direction is the Fisher linear discriminant [32, 33]. Fisher LDA seeks the direction w that maximises the ratio of between-class to within-class variance,

$$J(w) = \frac{\left(w^\top(\mu_{c_1} - \mu_{c_0})\right)^2}{w^\top S_W\, w}, \tag{5}$$

where $S_W^{(\ell)}$ is the pooled within-class covariance,

$$S_W^{(\ell)} = \frac{1}{|D| - 2} \sum_{c \in \{c_0, c_1\}} \sum_{x \in D_c} \big(h^{(\ell)}(x) - \mu_c^{(\ell)}\big)\big(h^{(\ell)}(x) - \mu_c^{(\ell)}\big)^\top. \tag{6}$$

The maximiser of (5) is $w \propto S_W^{-1}\Delta\mu$, which we compute with the regularised covariance Σ_s,

$$w_{\mathrm{LDA}}^{(\ell)} \propto \big(\Sigma_s^{(\ell)}\big)^{-1}\Delta\mu^{(\ell)}, \tag{7}$$

where $\Sigma_s^{(\ell)} = (1-s)S_W^{(\ell)} + s\ \mathrm{diag}\big(S_W^{(\ell)}\big)$. With $|D| = 538$ examples in $d = 2560$ dimensions, the within-class covariance is rank-deficient and not invertible. We therefore regularise it by shrinkage toward its diagonal, which shrinks the within-class correlations while keeping the per-feature variances, at a fixed shrinkage intensity $s = 0.1$ [34]. This intensity is fixed a priori for all analyses, and since the attenuation of Appendix E weakens monotonically as the intensity is raised, the reported orthogonality is not produced by the regularisation.

By (7), w_{LDA} is $\Delta\mu$ whitened by the inverse within-class covariance, the direction of maximal linear class separability. We call it the reading direction because it is the optimal linear readout of the class. Whether this readout coincides with the direction the model writes, $\Delta\mu$, or uses, u, is an empirical question addressed in Section 3.1. The normalised reading direction is

$$\widehat{w}_{\mathrm{LDA}}^{(\ell)} = \frac{\big(\Sigma_s^{(\ell)}\big)^{-1}\Delta\mu^{(\ell)}}{\big\|\big(\Sigma_s^{(\ell)}\big)^{-1}\Delta\mu^{(\ell)}\big\|}.$$

2.5.3 Use direction u

The use direction is obtained by differentiating the class margin with respect to the residual stream. We define the margin as the difference between the logits of the two class tokens,

$$\mathrm{margin}(x) = \mathrm{logit}_{c_1}(x) - \mathrm{logit}_{c_0}(x), \tag{8}$$

where $\mathrm{logit}_c(x)$ is the logit of the class c token at the answer position. The margin gradient with respect to the residual stream at layer ℓ points in the direction in which a local change of the residual stream most strongly increases the preference for c_1 over c_0 [18]. The class-averaged use direction is

$$u^{(\ell)} = \frac{1}{2}\left(\frac{1}{|D_{c_0}|}\sum_{x \in D_{c_0}} \nabla_{h^{(\ell)}}\mathrm{margin}(x) + \frac{1}{|D_{c_1}|}\sum_{x \in D_{c_1}} \nabla_{h^{(\ell)}}\mathrm{margin}(x)\right). \tag{9}$$

The balanced average over the two classes in (9) gives the direction to which the model's decision is on average most sensitive, regardless of which class is correct for a given example. The normalised use direction is

$$\widehat{u}^{(\ell)} = \frac{\overline{\nabla_{h^{(\ell)}}\,\mathrm{margin}}}{\big\|\overline{\nabla_{h^{(\ell)}}\,\mathrm{margin}}\big\|},$$

where the overbar denotes the class average, as in (9).

2.5.4 Unembedding direction γ

The unembedding direction is the contrast in the unembedding space between the class tokens,

$$\gamma = W_U[c_1, :] - W_U[c_0, :], \tag{10}$$

where $W_U[c, :]$ is the row of the unembedding matrix corresponding to the class c token [4]. Equivalently, γ is a direction in the residual stream space, and the projection of the final normalised state onto γ is the logit difference (8). We use γ as a reference for interpretation. The alignment of the use direction u with γ, and the non-alignment of the reading direction w_{LDA} with γ, are examined in Section 3.1. The normalised unembedding direction is $\widehat{\gamma} = \gamma/\|\gamma\|$.

Symbol	Name	Definition	Role
$\Delta\mu$	writing	difference of class means, (4)	content written into the stream
w_{LDA}	reading	Fisher discriminant, (7)	optimal linear readout
u	use	class average of the margin gradient, (9)	direction of strongest local effect on the output
γ	unembedding	unembedding contrast, (10)	output direction of the logit difference

Table 1: The four geometric directions, their definitions and roles.

2.6 Decomposition of the change of a direction

The change of a direction between adjacent layers is decomposed in two complementary ways. Geometrically, for an increment $\delta v = v^{(\ell)} - v^{(\ell-1)}$, it splits into a length component along the current direction and a rotation component orthogonal to it (Appendix C),

$$\mathrm{len}(\delta v) = \delta v^{\top} \widehat{v}^{(\ell-1)}, \qquad \mathrm{rot}(\delta v) = \left\| \delta v - \mathrm{len}(\delta v)\, \widehat{v}^{(\ell-1)} \right\|, \tag{11}$$

separating changes that rescale a representation from those that reorient it. Mechanically, the additive structure of the residual stream (1) splits the change of the writing direction into an attention and an MLP contribution. Since $\Delta\mu$ is a linear function of the class means, which inherit that additivity,

$$\Delta\mu^{(\ell)} - \Delta\mu^{(\ell-1)} = \underbrace{\left(\bar{a}^{(\ell)}_{c_1} - \bar{a}^{(\ell)}_{c_0}\right)}_{\text{attention}} + \underbrace{\left(\bar{m}^{(\ell)}_{c_1} - \bar{m}^{(\ell)}_{c_0}\right)}_{\text{MLP}}, \tag{12}$$

with $\bar{a}^{(\ell)}_c$ and $\bar{m}^{(\ell)}_c$ the class-averaged attention and MLP contributions of layer ℓ. The identity holds to machine precision and uses the true MLP output captured on the forward pass, not its transcoder reconstruction, which is unreliable on the small class difference. Attributing the rotation of (11) to each block through (12) gives the MLP share of the rotation,

$$\mathrm{MLP\%_{rot}} = \frac{\left\langle |\mathrm{rot_{MLP}}| \right\rangle}{\left\langle |\mathrm{rot_{MLP}}| + |\mathrm{rot_{attn}}| \right\rangle}, \tag{13}$$

where $\mathrm{rot}_B(\delta v) = \langle \delta v_B, \widehat{r} \rangle$ is the contribution of block $B \in \{\mathrm{MLP}, \mathrm{attn}\}$ projected onto the unit rotation direction $\widehat{r} = \left(\delta v - \mathrm{len}(\delta v)\, \widehat{v}^{(\ell-1)}\right) / \mathrm{rot}(\delta v)$, with δv_{MLP} and δv_{attn} the two terms of (12). The signed projections are additive, $\mathrm{rot_{MLP}} + \mathrm{rot_{attn}} = \mathrm{rot}(\delta v)$, so the share in (13) is taken over the summed absolute block projections, which equal the total rotation when the two blocks turn the direction in the same sense. The share is averaged over the layers where the rotation is a substantial part of the change, the rotation fraction $\mathrm{rot}(\delta v)/\|\delta v\|$ being above 0.3, which excludes the layers where the direction barely moves and the attribution is unstable. The length share is defined in the same way. The decomposition (12) is mechanically exact only for the writing direction, which is linear in the residual stream. The use and reading directions are nonlinear functions of the activations, so for them the same split is a geometric alignment rather than a mechanical decomposition, and we claim MLP formation only for $\Delta\mu$ (Section 3.2).

2.7 Direction interventions

We assess the causal role of a direction by activation intervention [13, 14, 15]. At layer ℓ, at the answer position, the last token, we add a single direction to the residual stream,

$$h^{(\ell)} \leftarrow h^{(\ell)} + s\, c\, \sigma^{(\ell)}\, \widehat{v}, \qquad \widehat{v} \in \left\{ \widehat{\Delta\mu}^{(\ell)},\ \widehat{u}^{(\ell)},\ \widehat{w}^{(\ell)}_{\mathrm{LDA}} \right\}, \tag{14}$$

where $\widehat{v}$ is a unit direction, $s \in \{-1, +1\}$ sets the sign toward the target class, $c \in \{1, 4, 8, 16, 32\}$ is the strength, and $\sigma^{(\ell)}$ is the standard deviation of the activation projections at layer ℓ [15]. Because

the scale of the residual stream grows by orders of magnitude with depth, $\sigma^{(\ell)}$ is computed per layer, so that a given c is a comparable perturbation at every layer. The same $\sigma^{(\ell)}$ is used for all three directions, so the comparison reflects the direction rather than the size of the perturbation. The intervention is applied separately for each direction at every layer $\ell \in \{1, \ldots, 36\}$, giving a causal profile of each direction across layers. After the intervention the forward pass continues through the remaining layers, and the logits are read at the answer position.

2.8 Feature attributions

We measure the contribution of each feature to the answer by gradient times activation [25, 20]. For prompt p and feature k at layer ℓ,

$$\tau_k^{(\ell)}(p) = a_k^{(\ell)}(p)\, \langle W_{\text{dec}}^{(\ell)}[k],\, g^{(\ell)}(p) \rangle, \qquad g^{(\ell)}(p) = \frac{\partial\, \text{margin}(p)}{\partial\, m^{(\ell)}}, \tag{15}$$

where $a_k^{(\ell)}(p)$ is the activation of feature k, $W_{\text{dec}}^{(\ell)}[k]$ is its decoder direction, $m^{(\ell)}$ is the MLP output, and $g^{(\ell)}(p)$ is the gradient of the margin with respect to the MLP output. The inner product $\langle W_{\text{dec}}^{(\ell)}[k], g^{(\ell)}(p) \rangle$ is the effect on the margin of the content the feature writes, so attribution combines how strongly a feature activates with the first-order effect of its written direction on the margin.

Attribution separates presence from use. A feature can activate strongly yet have near-zero attribution if its decoder direction is orthogonal to the gradient, in which case it is present in the activations but does not move the margin to first order. Activation therefore measures presence, and attribution measures use.

We also use attribution to select the features for the rest of the analysis. The features with the largest attribution form an attribution graph, and these are the features we ablate (Se -ction 2.9) and group into clusters (Section 2.10).

2.9 Feature ablation

The causal role of a feature or of a cluster of features is tested by ablation, removing it and measuring the change in the output [25]. We subtract the feature's contribution $a_f^{(\ell)}(p)\, W_{\text{dec}}^{(\ell)}[:, f]$ from the true MLP output, so the intervention does not pass through the transcoder reconstruction, and the forward pass continues through the remaining layers. The effect on prompt p is the signed change in the margin,

$$e(f, p) = \text{margin}_{\text{ablated}}(p) - \text{margin}_{\text{baseline}}(p), \tag{16}$$

and we summarise a feature by the mean absolute effect over prompts and by the relative effect $|e|/|\text{margin}_{\text{baseline}}|$. These per-feature effects are the input to the feature clustering (Section 2.10). The geometric effect of ablation on the writing and use directions is reported in Appendix D.

2.10 Cluster formation

Because a single feature is polysemantic [26], we group features by their joint causal importance. For each feature we take the ten prompts on which its ablation has the largest absolute effect (Section 2.9), and define the co-importance of two features as the Jaccard overlap of these prompt sets,

$$\text{coimp}(i, j) = \frac{|T_i \cap T_j|}{|T_i \cup T_j|}, \tag{17}$$

where T_i is the set of the ten prompts with the largest ablation effect for feature i. We convert co-importance into a distance $D_{ij} = 1 - \text{coimp}(i, j)$ and cluster the features by hierarchical agglomerative clustering with average linkage. The distance between two clusters A and B is the mean

pairwise distance between their features,

$$d(A, B) = \frac{1}{|A|\,|B|} \sum_{i \in A} \sum_{j \in B} D_{ij}, \tag{18}$$

and at each step the two clusters with the smallest $d(A, B)$ are merged, starting from each feature as its own cluster. We cut the resulting hierarchy at $K = 12$ clusters, the value selected by a test of cluster stability under paraphrasing of the input (Appendix D), to obtain the clusters used as the working unit in Section 3.6.

3 Results

3.1 Geometry of directions

3.1.1 Decodability

The reading direction w_{LDA} (7) decodes the decay type almost perfectly, with a held-out AUC reaching 0.99 from the middle layers onward (Table 2). The writing direction $\Delta\mu$ is also decodable, with an AUC up to 0.96, so high decodability is not specific to the reading direction [1, 5]. The use direction u, by contrast, stays near chance in the middle layers and becomes decodable only near the output, a contrast taken up in Section 3.1.2.

layer	w_{LDA}	$\Delta\mu$	u
4	0.938	0.784	0.638
12	0.967	0.870	0.594
18	0.994	0.955	0.506
22	0.999	0.965	0.516
24	0.998	0.962	0.700
28	0.998	0.943	0.623
32	0.998	0.915	0.718
35	0.993	0.917	0.877

Table 2: Held-out AUC of the per-layer projection onto each direction, α/β task, held-out 50/50 split.

3.1.2 Causal inertness

Despite high decodability, intervention along w_{LDA} leaves the answer almost unchanged on the α/β task. We measure the causal effect by the margin flip, the fraction of examples whose class margin reverses sign,

$$\text{margin flip} = \mathbb{E}_x\big[\ \text{sign margin}'(x) \neq \text{sign margin}(x)\ \big], \tag{19}$$

where $\text{margin}(x)$ is the quantity of Equation 8 and $\text{margin}'(x)$ is its value after the intervention. We use the class margin rather than the top token, because the top token is format-dependent and undercounts flips that produce coherent text rather than a bare class token.

At strength $c = 16$ the causal effect of w_{LDA} depends on the concept and tracks its geometry (Table 3). On α/β the reading direction is approximately orthogonal to the use direction ($\cos = 0.02$) and causally inert, with a margin flip of 0.07 against a random control of 0.01 (a random unit direction at the same strength). On the particle concept the reading direction is non-orthogonal to use ($\cos = 0.15$) and causal, with a margin flip of 1.00 against a control of 0.23. The writing direction, non-orthogonal to u on both concepts, and the use direction itself flip the margin in both cases. The reading direction is therefore causally inert exactly where it is orthogonal to the use direction and causal where it aligns with it, so the causal effect of the reading direction is mediated by its component along the use direction rather than by the reading direction itself. This concept dependence is not an artefact of the strong intervention. At $c = 8$, where the random control is still near zero, the reading direction flips 0.48 on the particles against 0.00 on α/β, and it stays below 0.02 on α/β at every strength, so the causal effect of reading on the particles tracks its alignment with the use direction rather than saturation at large strength.

The dissociation is symmetric. The reading direction decodes the class almost perfectly yet leaves the answer unchanged, while the projection onto the use direction decodes barely above

chance in the decision zone (AUC 0.516 at layer 22, Table 2) at the very layers where intervention along it reverses the margin on 0.88 of the examples (Table 3). One direction carries the class information without causal effect, and the other carries the causal effect without linearly readable class information.

concept	$\cos(w_{\text{LDA}}, u)$	$\cos(\Delta\mu, u)$	w_{LDA}	$\Delta\mu$	u	random
α/β	0.02	0.34	0.07	0.88	0.88	0.01
particles	0.15	0.26	1.00	1.00	1.00	0.23

Table 3: Geometry and causal effect of the three directions for two concepts. The last four columns are margin flips.

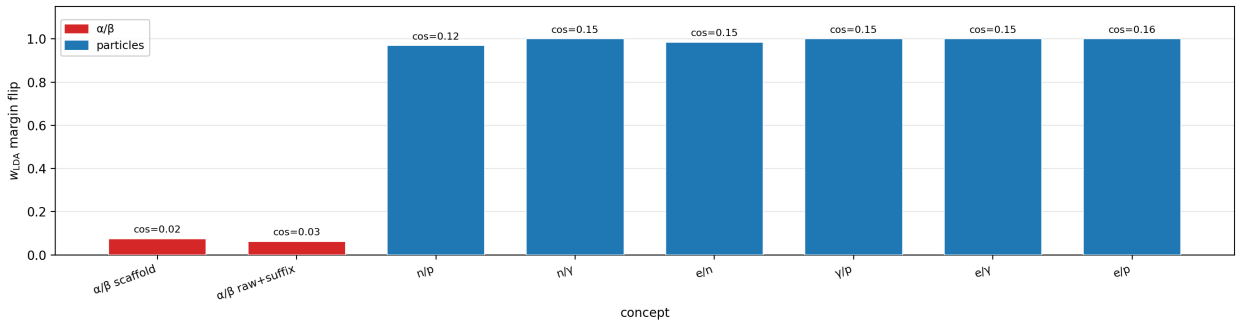


Figure 1: Causal effect (margin flip) of the reading direction against its alignment with the use direction, across concepts.

3.1.3 Orthogonality and output alignment

The reading and use directions are approximately orthogonal, with a cosine averaged over layers of $\cos(w_{\text{LDA}}, u) \approx 0.02$. This orthogonality is not an empirical accident but a consequence of an exact attenuation identity (Appendix E). The whitening in Equation 7 multiplies the alignment of the reading direction with any axis of the covariance by the ratio of λ_{eff}, the dispersion scale that $\Delta\mu$ occupies after whitening, to the within-class dispersion along that axis. The use direction is a high dispersion axis, since the within-class fluctuations of the margin lie along it, so the optimal linear probe must turn away from u. Because a small cosine is expected in high dimension, this orthogonality is assessed against a covariance-matched null and an attenuation bound rather than the isotropic baseline (Appendix A).

This is confirmed by the alignment with the unembedding contrast γ (Table 4). With depth the use direction converges to γ, reaching $\cos(u, \gamma) = 0.99$ at the final layer. The writing direction aligns partially, its cosine with γ peaking at 0.42 and sitting at 0.33 at the final layer, and the reading direction stays orthogonal to γ, with $\cos(w_{\text{LDA}}, \gamma) \approx 0.01$. Because the projection onto γ is the logit difference (8), the alignment of u with γ and the orthogonality of w_{LDA} to γ are independent evidence that the concept enters the prediction through u, not through w_{LDA}.

The geometry across layers matches the causal effect (Table 4). The reading direction stays orthogonal to both u and γ at every layer and its margin flip never exceeds 0.07. The use direction aligns with γ first and is causal earliest, with a margin flip of 0.41 at layer 20. The writing direction aligns with use around layers 21 to 22 and becomes causal there, with a margin flip of 0.85 at layer 22. The directions aligned with γ, use and writing, are causal, while the reading direction, orthogonal to it, is inert. These onsets fall within a narrow band of layers, the decision zone centred on L21, where a set of independent measurements first exceed their nulls (Appendix B).

Because u is the gradient of the margin, a first-order step along u raises the margin by construction, so this identity is not itself a finding. Three of the results above do not follow from it. The reversals are measured at a finite step of 16σ (Table 3), realised through the full nonlinear network

L	$\cos(w,u)$	$\cos(\Delta\mu,u)$	$\cos(u,\gamma)$	$\cos(w,\gamma)$	$\cos(\Delta\mu,\gamma)$	w	$\Delta\mu$	u
16	0.03	0.06	0.04	0.03	0.00	0.00	0.00	0.05
20	0.02	0.09	0.05	0.03	0.01	0.03	0.12	0.41
22	0.03	0.29	0.14	0.01	0.02	0.06	0.85	0.88
24	0.00	0.40	0.27	0.01	0.14	0.03	0.88	0.88
28	0.03	0.39	0.47	0.02	0.15	0.05	0.88	0.88
32	0.01	0.42	0.64	0.01	0.42	0.03	0.88	0.88
35	0.03	0.34	0.99	0.03	0.33	0.01	0.88	0.88

Table 4: Per-layer cosines and margin flip ($c = 16$), α/β. The last three columns are margin flips of w_{LDA}, $\Delta\mu$, and u.

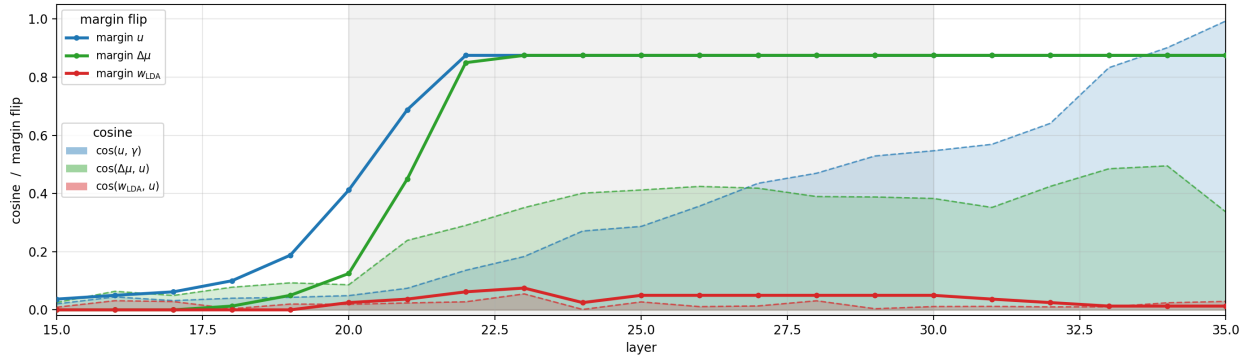


Figure 2: Alignment of the three directions with the use direction and the unembedding contrast across layers, together with the causal margin flip.

rather than the first-order term. The alignment of u with γ forms within the narrow decision zone at L21 rather than uniformly with depth (Table 4). The equality $u = \gamma$ at the readout is definitional, but the depth at which the alignment forms is not. The reading direction w_{LDA} is defined from the class covariance (7), is independent of u, decodes the class at AUC 0.99 (Table 2), and yet its margin flip does not exceed 0.07 (Table 3), which no gradient identity forces. The contrast between the direction that reads the class and the direction that produces it is the finding, and the definition of u does not supply it.

3.2 MLP blocks and writing direction

Decomposing the rotation of the writing direction into blocks on the true MLP output (Section 2.6) shows that the MLP blocks carry most of it, and the result is invariant in two independent ways (Table 5). Across the three framings of the α/β task the MLP share of the rotation stays above one half, from 56 percent on the raw form with suffix to 70 percent on the raw form, so the mechanism tracks the concept rather than the surface form. Across the six particle pairs it lies in the narrow band 82 to 86 percent, so it is not tied to one pair either. The MLP share of the rotation across the pairs is far more tightly grouped than its share of the length (42 to 62 percent), so the rotation is the purer MLP effect while attention contributes more to the length.

The MLP contribution shifts with depth (Figure 3). In the early layers the MLP carries more of the length of the writing direction than of its rotation, and in the late layers more of the rotation, a crossover present on every concept. The late layers, where the rotation of the writing direction is most MLP driven, are also where the writing direction is formed, 91 percent of it in layers 27 to 35 (Section 3.5). The block decomposition and the feature capture of Section 3.5 are therefore two views of the same late formation, the MLP blocks that form the writing direction and the transcoder features that capture it above the random null.

The decomposition uses the true MLP output (Section 2.6). Within the particle pairs the MLP share varies little, from 82 to 86 percent, so the small spread carries no evident structure. The gap

between the two bands, 56 to 70 percent on the α/β framings against 82 to 86 percent on the particle pairs, is left open, the two concepts differing in corpus size and in prompt structure.

concept	rotation	length
α/β raw	70	66
α/β scaffold	60	40
α/β raw with suffix	56	52
electron/photon	86	62
electron/neutron	85	53
electron/proton	85	55
neutron/photon	84	53
photon/proton	85	52
neutron/proton	82	42

Table 5: MLP share of the rotation and length of the writing direction $\Delta\mu$ (Equation 13, averaged over layers with rotation fraction above 0.3), computed on the true MLP output.

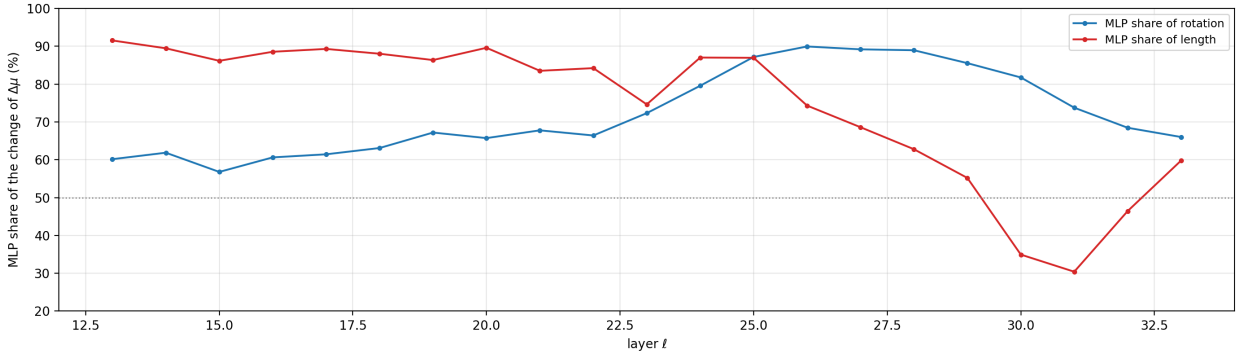


Figure 3: MLP share of the length and of the rotation of the writing direction $\Delta\mu$ across layers, on the α/β decay corpus. The MLP carries the length early and the rotation late, the two crossing near the end of the decision zone.

3.3 Dissociation under intervention

The three directions differ in their effect on the generation under intervention. The writing direction $\Delta\mu$ and the use direction u flip the class margin and are causal (Section 3.1.2), while the reading direction w_{LDA} is inert at moderate strength and moves the margin only at large strength, through its small component along the use direction (Section 3.1.2). Their effect on the generation still differs. We steer at the answer position of a late layer (L24) on the particle prompts and read the continuation as the strength c grows (Tables 6 and 7).

The use direction forces the class token and, as the strength grows, degenerates the continuation into a repetition of it, because it aligns with the unembedding contrast γ toward the output (Section 3.1.3), the direction that sets the class token logits. The writing and reading directions are orthogonal to γ and write content into the residual stream rather than raising a single token logit, so their continuations stay coherent rather than collapsing into a repeated token. Of the three the reading direction changes the answer last, holding the correct class to a higher strength than $\Delta\mu$ and u before it too flips at the largest strength, its margin on the particles turning over as well (Table 3). A top-token metric would miss this content, which is why the comparison uses the class margin and the generation rather than the top token.

Each steering direction is a fixed vector, the between-class contrast estimated on held-out prompts of the pair, and adding it moves the answer to the paired particle rather than to an arbitrary class, even though the prompt does not mention it. This specificity is consistent with the categorical

steered	c	generation
$\Delta\mu$	4	The neutron. The number of protons in the nucleus determines the element. The number of neutrons determines the isotope
$\Delta\mu$	8	The proton is a positively charged particle, and when added to the nucleus it increases the atomic number of the element, thereby changing the element
$\Delta\mu$	16	The proton is a positively charged particle with a charge of $+1$ and a mass of $+1$. The element is defined by the number of protons
u	4	The proton is a positively charged particle with a charge of $+1e$ and a mass of 1.6726231
u	8	Protons are positively charged, and electrons are negatively charged. The number of protons in the nucleus is equal to the number of electrons
u	16	Proton, proton, proton, proton, proton, proton, proton, proton, proton
w_{LDA}	4	Neutrons. The number of neutrons in the nucleus of an atom is called the neutron number
w_{LDA}	8	The neutron. The proton is a positively charged particle, and the neutron is a neutral particle
w_{LDA}	16	The proton is a positively charged particle, and when added to the nucleus it increases the atomic number of the element

Table 6: Generations under steering on a particle prompt as the strength grows (correct answer neutron). The prompt is "Unlike the proton, which adds both charge and mass to the nucleus, which neutral nuclear particle can be added without changing the element?".

steered	c	generation
$\Delta\mu$	4	Photons of ultraviolet light have enough energy to eject electrons from the zinc plate. The ejected electrons are released as photons
$\Delta\mu$	8	Photons of ultraviolet light have a higher energy than photons of visible light. The energy of a photon is given by the equation $E = h\nu$
$\Delta\mu$	16	Photons of the same energy as the photon of the same energy as the photon of the same energy as the photon
u	4	Photon. The energy of the photon is transferred to a photon of light, which is then absorbed by the zinc plate
u	8	Photon. The photon is a particle of light that carries a photon of energy. The photon is a particle of light
u	16	photon photon photon photon photon photon photon photon photon
w_{LDA}	4	Electromagnetic radiation is a form of energy that travels through space as a wave. It includes visible light, radio waves, microwaves
w_{LDA}	8	Photons of ultraviolet light are absorbed by the zinc plate, causing the release of photons of light
w_{LDA}	16	Photons of electromagnetic radiation are the particles that are released. What is the difference between a photon and a particle of electromagnetic radiation?

Table 7: Generations under steering on a second particle prompt as the strength grows (correct answer electron). The prompt is "When ultraviolet light hits a zinc plate, negatively charged particles are ejected in the photoelectric effect. Which particles are released?".

geometry of Park et al. [35]. In their account the classes of a categorical concept form a simplex in the representation space, and the between-class contrast is the edge between two vertices, so a push along $\Delta\mu$ moves the state along that edge toward the neighbouring vertex and not toward the other classes.

3.4 Comparison of direction norms

Sections 3.1 and 3.3 separate the writing and use directions by their causal role. This section adds a geometric description. Relative to the norm of the residual stream, the two directions behave oppositely with depth, which gives a mechanistic reading of the content versus token dissociation of Section 3.3. We measure the ratio of each direction's norm to the residual stream norm by layer,

$$r_v^{(\ell)} = \frac{\|v^{(\ell)}\|}{\|h^{(\ell)}\|}, \qquad v \in \{\Delta\mu, u\},$$

comparing the early layers ($\ell < 22$) with the later layers ($\ell \geq 22$), for several concepts (Table 8). The writing direction $\Delta\mu$ grows relative to the stream toward the output, for example from 0.05 to 0.35 on the scaffold and from 0.16 to 0.47 on the particles, while the use direction u shrinks. The growth of the writing direction is framing-dependent and is weaker on the raw form with suffix (0.024 to 0.14) than on the scaffold, but the shrinking of the use direction is present on every concept tested. The gap between the two norms reaches between two and three orders of magnitude toward the output, where $\|\Delta\mu\|$ is up to about 900 times $\|u\|$.

concept	$r_{\Delta\mu}$ early→zone	r_u early→zone	$\|\Delta\mu\|/\|u\|$ (L35)
α/β scaffold	$0.05 \rightarrow 0.35$	$0.023 \rightarrow 0.003$	726
α/β raw with suffix	$0.024 \rightarrow 0.14$	$0.075 \rightarrow 0.004$	234
particles (e/γ)	$0.16 \rightarrow 0.47$	$0.013 \rightarrow 0.003$	903

Table 8: Norm of the writing and use directions relative to the residual stream, from the early layers to the later layers, and their ratio at the last layer.

This gives a mechanistic reading of the dissociation. The between-class difference is written into the stream as content. Toward the output it grows in norm and takes up an increasing share of the stream, because the between-class difference becomes large and embedded. The use direction requires not a large norm but a precise orientation, so it shrinks relative to the stream while remaining the direction to which the output is sensitive (Section 3.1.3). The writing direction grows into the stream, the use direction shrinks within it.

This geometry separates the two directions in a way their rotation does not. As shown in Appendix C, both rotate by a similar angle across depth, the common background rotation of the task subspace, so it is not the rotation that distinguishes them but their relation to the stream norm.

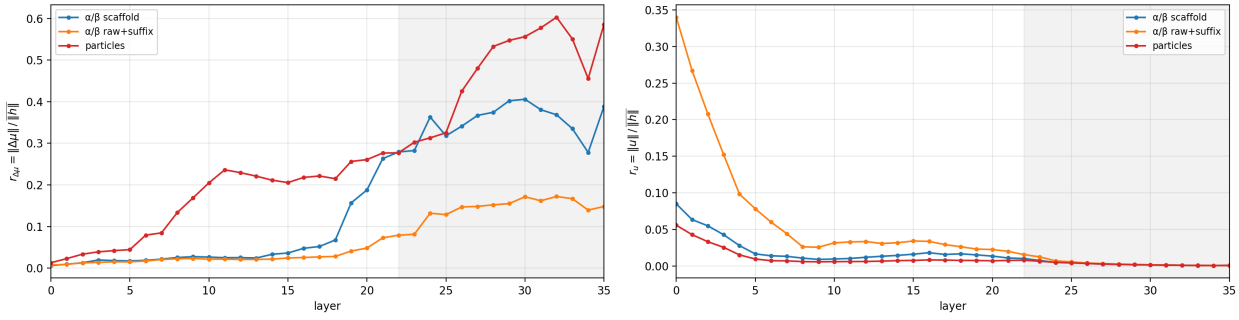


Figure 4: Norm of the writing and use directions relative to the residual stream norm, across layers.

3.5 Feature span projection on the directions

The reading, writing, and use directions differ in whether the transcoder features can represent them. We restrict to the active subdictionary A, the features active on at least 5 percent of prompts, and measure how much of each unit direction lies in the span of their decoder directions,

$$\text{capture}(v) = \frac{\|P_{\text{span}(D_A)}\,\hat{v}\|^2}{\|\hat{v}\|^2}, \qquad D_A = \{\, W_{\text{dec}}[f] : f \in A^{(\ell)} \,\},$$

where P is the orthogonal projection onto the column space of D_A. Capture is 1 when $\hat{v}$ lies entirely in the active span and 0 when it is orthogonal to it. The random null is the same quantity for a random unit vector, which in expectation equals $|A|/d$, the fraction of any direction that an $|A|$-dimensional subspace captures. Here it is about 0.015 to 0.035, matching $|A| \approx 40$ to 90 out of $d = 2560$ (Table 9).

layers	$\lvert A\rvert$	$\Delta\mu$	u	w_{LDA}	null	$\Delta\mu$/null	u/null	w/null
early (8–15)	38	0.054	0.021	0.019	0.015	3.6	1.4	1.3
middle (20–26)	87	0.207	0.077	0.045	0.035	5.9	2.2	1.3
late (30–35)	43	0.242	0.044	0.027	0.016	15.1	2.7	1.7

Table 9: Active feature span capture by depth band. $|A|$ is the number of active features, with the capture of the writing direction $\Delta\mu$, the use direction u, and the reading direction w_{LDA} against a random null ($\approx |A|/d$, $d = 2560$) and the ratios to the null. The three rows are coarse depth groupings for the capture trend, not the decision zone proper (L19 to L24, Appendix B).

The writing direction is captured well above the null, by a factor of about 6 in the middle band (capture 0.207 against a null of 0.035) and 15 in the late layers. The use direction is captured partially, by about 2.2. The reading direction remains within a factor of 1.3 to 1.7 of the null across bands, far below the factors 6 to 15 of the writing direction. What the features write lies within the reach of the dictionary, while the optimal reading direction does not. This is the feature-level counterpart of the geometry of Section 3.1.3. The whitening rotates the reading direction out of the subspace in which the MLP writes, consistent with the capture at the null (Appendix E).

Capture is computed with free coefficients, so it establishes that a direction is expressible within the span of the active decoder directions, not that the features write it with their realised activations.

The absolute capture is modest, around 0.2 to 0.3 for the writing direction, because the active subdictionary is small, 10 to 135 features per layer, and the between-class difference partly lies outside the span of the features. The contrast with the null, not the absolute value, is the result.

The population of writing features is not static across depth. Each layer has its own dictionary, and the number of features that write the between-class difference expands in the middle band (L20 to L26), from about 8 in the early layers to 56 in layers 20 to 26, where the writing is also most distributed, the five strongest features accounting for 70 percent of the written difference against about 95 percent in the early and late layers. Most of the contribution to $\Delta\mu$ occurs in the late layers, 91 percent of it in layers 27 to 35, the same layers where the MLP blocks dominate the rotation of the writing direction (Section 3.2). The writing is a distributed process spread across depth, not produced by a fixed set of features.

3.6 Clustering

3.6.1 Cluster features and physical meaning

Features are grouped into clusters by their joint causal importance (Section 2.10), and each cluster is enriched for a specific physical feature of the decay. Enrichment is measured by the lift, the ratio of the frequency of a cue feature among the prompts on which the cluster has the largest causal effect

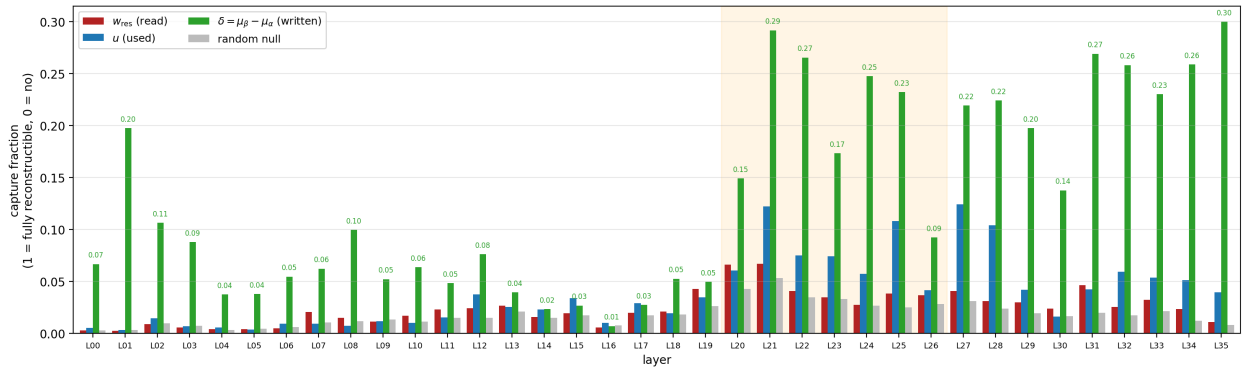


Figure 5: Active feature span. Capture of the reading, use, and writing directions from the active dictionary per layer, with a random null.

to its baseline frequency,

$$\text{lift}(C) = \frac{p(\text{cue} \mid \text{prompts where } C \text{ has the largest ablation effect})}{p(\text{cue})}. \tag{20}$$

A lift of 1 means no enrichment, the cue appearing among the cluster's most affected prompts no more often than at baseline, and a lift well above 1 means strong enrichment. All clusters are enriched for a physical feature with a lift of 6.3 to 18.8 (Table 10), and the features are physically meaningful and ordered by depth, with charge and lepton number in the early layers, nuclear composition and the type of transition in the middle, and the output decision in the late layers.

To determine whether the cluster features detect a physical feature or a class label, we measure their mean activation separately on the prompts of each class. For feature j and class c, the mean activation is

$$\text{fire}_j^{(c)} = \mathbb{E}_{x \in D_c}\left[f_j(x)\right], \tag{21}$$

the average activation of the feature over all prompts of class c, counting prompts on which it is inactive as zero. The class offset of a feature is the difference $\text{fire_diff}_j = \text{fire}_j^{(c_1)} - \text{fire}_j^{(c_0)}$, and Table 10 reports its mean absolute value over the features of each cluster. This activation offset is distinct from a frequency-based specificity, which compares firing frequencies rather than activation magnitudes. A feature that detects a class label activates strongly on one class and weakly on the other, so its class offset is large. A feature that detects a physical feature present in both classes activates comparably on both, so its offset is small relative to the activation level.

The cluster features activate almost equally on the two classes (Table 10). For the charge cluster the mean activation is 2.67 on the α class and 2.77 on the β class, for the lepton cluster 2.40 and 2.35, and for the transition cluster 1.94 and 2.03. The class offset is small relative to the activation level in every cluster, with the ratio of the mean absolute offset to the mean activation between 0.01 and 0.11. Charge features activate in both α and β contexts, because charge is a physical feature present in both types of decay. They detect charge, not class.

The role labels that the attribution graph assigns to features, such as α-attribution or β-attribution, reflect a contrast of activation weighted by the gradient rather than class-selective activation. A cluster can carry such an attribution role label while its features activate evenly across the two classes, which is exactly what the activation offset of Equation 21 measures directly.

3.6.2 Sequence of physical features

Combining the two facts established above, that each cluster rotates the writing direction on its own layer (Appendix D.4) and that each cluster detects a physical feature, the formation of the writing direction can be read as a sequence in depth (Figure 6). The physical features of the decay are inscribed in order. The conserved quantities enter in the early layers, charge at L10 and lepton

cluster	physical feature	lift	fire α	fire β	mean |offset|	layer	status
1	charge	11.8	2.67	2.77	0.12	L10	accepted
2	lepton family	11.8	2.40	2.35	0.10	L11	partial
3	Z+1 and antineutrino	17.6	1.82	1.83	0.01	L12	descriptive
4	charge -1	7.8	1.97	1.89	0.08	L13	accepted
5	n$\rightarrow$p transition (hub)	18.8	1.94	2.03	0.20	L14–22	partial
6	composite nuclear object	11.8	3.07	3.37	0.29	L15	descriptive
7	charge -1	9.8	0.49	0.46	0.04	L18	descriptive
8	composite nuclear object	17.6	1.36	1.31	0.12	L19	accepted
9	Z+1 and antineutrino	11.8	2.17	1.94	0.23	L20	partial
10	heavy nuclear fragment	15.7	3.06	2.92	0.14	L21	accepted
11	composite nuclear object	11.8	7.75	7.80	0.05	L23	accepted
12	Z+1 output decision	6.3	5.27	5.09	0.39	L24–25	accepted

Table 10: Clusters of the α/β task (HAC $K = 12$), ordered by depth. Lift is Equation 20, the per-class activations Equation 21. The status labels (accepted, partial, descriptive) and the criteria behind them are described in Appendix D.

number at L11. The nuclear composition and the type of transition enter in the middle layers, the neutron-to-proton transition at the L14 to L22 hub and the composition of the nuclear fragment across L15 to L23. The output decision is inscribed in the late layers at L24 to L25. In other words the writing direction does not carry the label α or β. The features write a physical description of the process into the residual stream, which charge, which lepton number, which nuclear composition, and which type of transition, and the class label is assembled from this description in the later layers.

This gives a mechanistic reading of the rotation of the writing direction with depth (Section 3.2) that complements the feature-level reading of Section 3.5. The writing direction rotates across layers because different clusters are active at different layers and inscribe different physical features. An early rotation carries the charge feature, the next the lepton number, and so on toward the output. The rotation of the writing direction is therefore not the accumulation of a class signal but a sequence of physical features being written, and the conceptual α or β decision is a late assembly from this physical description.

3.6.3 Structural detection

If the early clusters detect physical features rather than classes, then asking which class the charge cluster writes toward is ill-posed at the level of the representation. Charge is neither α nor β, it is a feature present in both types of decay, and the cluster activates evenly on both. The absence of a pure class signal in the early writing is therefore not a defect of the measurement but a consequence of the early writing encoding the physics of the process rather than the class label.

This is consistent with two earlier results. The use direction, which carries the class decision, is separated from the early writing and assembled near the output (Sections 3.1.3 and 3.5). The attribution role of a feature does not predict the direction of its causal effect (Section 3.6, Appendix D). Both reflect the same fact. The class label is not present directly in the early layers. It is computed from the physical features and resolved near the output.

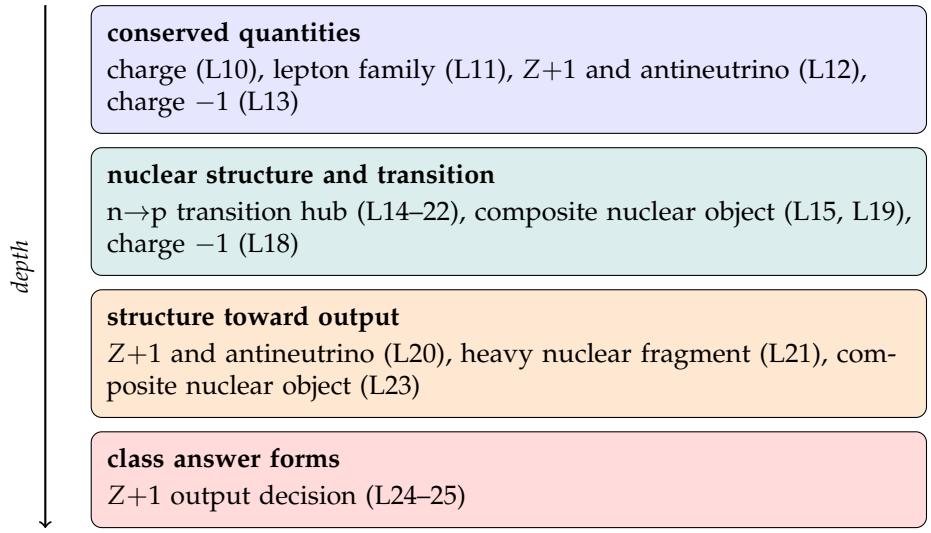


Figure 6: The α/β decay-type circuit organised by depth (HAC $K = 12$ clusters). Each cluster detects a physical feature of the decay with balanced activation on the two classes, and the features enter in order of depth, the conserved quantities first, then the nuclear structure and transition type, and the class answer only in the late layers. Cluster indices follow Table 10, and the per-cluster rotation angles are reported in Appendix D.4.

4 Discussion & Conclusion

4.1 Discussion

The results separate the single notion of a concept direction into three causally and mechanically distinct objects. The reading direction w_{LDA} decodes the concept almost perfectly, with an AUC of 0.98 to 0.99, yet on the α/β task intervention along it leaves the answer unchanged. The inertness is geometric rather than universal. On the particle pairs, where the reading direction has a larger component along the use direction, the same intervention flips the answer, so causal relevance tracks alignment with u rather than decodability. The use direction is causal on every concept tested, and with depth it aligns with the unembedding contrast γ, through which the concept reaches the prediction.

The dissociation is symmetric. The reading direction carries the class information without causal effect, and the projection onto the use direction decodes barely above chance in the decision zone at the layers where intervention along it reverses the margin on most examples. The within-class dispersion along u links the two halves. It masks the between-class separation along u from a linear reader, and by the attenuation identity of Appendix E it also attenuates the optimal probe along u, so the probe avoids the direction to which the output is sensitive for the same reason that the projection onto that direction decodes poorly.

Under intervention the writing and use directions dissociate qualitatively. Steering along the writing direction moves the continuation toward a coherent description of the target concept, whereas steering along the use direction raises the target token while the continuation degenerates into repetition. The two directions are also opposite relative to the stream, the writing direction growing in norm on the scaffolded framings while the use direction shrinks on all of them. The specificity of the shift, toward the paired class rather than an arbitrary one, is consistent with the categorical geometry of Park et al. [35].

The writing direction is formed by the MLP blocks and lies in the span of the active transcoder features well above a random null, so the mechanism that builds it and the dictionary that expresses it describe one process. The features detect the physical properties of the decay, the charge, the lepton number, the nuclear composition, and the transition type, rather than the class, activating comparably on both classes. The model builds a physical description of the process and resolves the class in the later layers, which places the task within the view that an abstract category is computed from more elementary features rather than represented directly [21].

Three reservations bound these claims. The use direction is a first-order object, and although the reported reversals are realised through the full nonlinear network, its interpretation beyond the local regime is not guaranteed. The span capture is computed with free coefficients and establishes expressibility by the active dictionary rather than a realised write. All results concern one model and two concept families, so the generality of the trichotomy remains an empirical question.

4.2 Conclusion

Linear probes establish that a concept is decodable from the residual stream, not that the decoded direction takes part in the computation. On a controlled binary task we showed that the single notion of a concept direction separates into three measured objects. The reading direction decodes the class almost perfectly yet is causally inert exactly where it is orthogonal to the use direction. The writing direction, formed by the MLP blocks and captured by the transcoder features above a random null, carries the content of the concept into the residual stream and grows into it with depth. The use direction is causally effective, barely decodable in the decision zone, compressed in norm, and aligned with the unembedding toward the output. Each direction therefore answers a different question, where the class can be read, what the model writes, and what moves the answer,

and none of them substitutes for the others.

Tracing the geometry through the transcoder features showed that the written content is a distributed combination of physical properties of the process, the charge, the lepton number, the nuclear composition, and the transition type, from which the class is assembled in the later layers. For interpretability the consequence is twofold. Decodability, whether of a direction or of a feature, must be paired with a causal test before it is identified with a mechanism, and the content a model writes need not be the category it predicts but the more elementary properties from which that category is computed. Further work includes testing the trichotomy on other model families, replacing the linear reader by nonlinear probes, measuring the realised activation-weighted write of the features, and extending the tasks beyond binary classification.

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A Orthogonality in high dimension

A small cosine between two directions is not on its own evidence of orthogonality, because in high dimension almost any two directions are close to orthogonal. For isotropically distributed unit directions in $\mathbb{R}^d$ the expected absolute cosine is

$$\mathbb{E}|\cos| \approx \sqrt{\tfrac{2}{\pi d}}, \tag{22}$$

which for $d = 2560$ is about 0.016. A small cosine is therefore expected under isotropy and is not by itself evidence of a structured relationship. Real representations are anisotropic, so the isotropic baseline is not the appropriate null, and we compare the measured cosine not against zero or the isotropic baseline but against a null distribution of cosines matched to the covariance of the activations, which accounts for the anisotropy of the representation [36, 37].

The effective dimension of the representation is estimated from the participation ratio of the eigenvalues of the activation covariance,

$$d_{\text{eff}} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{23}$$

Its value lies well below the nominal $d = 2560$, of order 13 in the decision layers and between about 3 and 20 across all layers. Against this covariance-matched null the reading and use directions are orthogonal. The measured $\cos(w_{\text{LDA}}, u)$ lies at about the 18th percentile of the null, below its median, and does not exceed the 95th percentile on any of the 36 layers. The measured cosine would exceed the isotropic threshold, but this reflects the inadequacy of the isotropic null under anisotropy rather than an alignment of the two directions.

Because the reading direction is estimated with finite reliability ($r_w \approx 0.72$ to 0.78, against $r_u \approx 0.99$ for the use direction), we apply the standard correction for attenuation, which gives an upper bound on the true correlation,

$$\rho_{\text{true}} \leq \frac{\rho_{\text{obs}}}{\sqrt{r_w r_u}}. \tag{24}$$

The per-layer upper bound on the correlation between reading and use never exceeds 0.074, and at the typical cosine it is a few hundredths. This bound is the main quantitative statement of the orthogonality, and the primary evidence that reading and use are distinct directions is the causal dissociation of Section 3.1.2, for which the geometric orthogonality is corroboration. For every reported cosine the measured value, the null median, the significance threshold (95th percentile), the null type, and the effective dimension are stated, so that each orthogonality is evaluated against the anisotropy of the representation rather than the isotropic baseline.

B Decision zone

The reading, writing, and use directions all change most sharply within a narrow band of layers rather than uniformly across depth. This appendix collects the evidence that the model commits to its answer within a specific band, which we call the decision zone, and locates its centre.

We take a set of measurements from different families, geometric, dictionary, feature, attribution, and causal. Each of them rises gradually with depth, and for each we record the layer at which it first exceeds its null. The null of a measurement is the value it takes under a matched random control, a random direction or a random set of features of the same size. Table 11 lists the measurements together with what each one shows.

measurement	family	onset	value
the two classes pull apart along the use direction by more than 1σ	geometric	L19	1.72σ
a linear classifier on the residual reaches AUC 0.96	geometric	L21	AUC 0.96
the writing direction enters the span of the active dictionary features	dictionary	L21	0.29 capture
the use direction enters the span of the active dictionary features	dictionary	L21	0.12 capture
dictionary features aligned with the use direction rise above the null	feature	L21	11 to 30
the attribution over the active features exceeds a matched random set	attribution	L21	$1.10\times$ null
steering along the use direction makes the target class token the top prediction	causal	L21–24	full

Table 11: Layer at which each measurement first exceeds its null, on the α/β decay corpus.

The measurements rise through the band from L19 to L24, and most of them reach their level at L21. The two classes begin to pull apart along the use direction (Section 2.5.3) at L19. By L21 the remaining measurements have risen above their nulls. A linear classifier on the residual reaches an AUC of 0.96, the full decodability of Section 3.1, whereas the projection onto the use direction alone remains close to chance (Table 2). The classes are therefore decodable from the residual without that decodability lying along the use direction. The writing and use directions come to lie in the span of the active dictionary features (the capture of Section 3.5), the number of dictionary features that write along the use direction grows, and the attribution over the active features rises above its random baseline (Section 2.8). The causal effect grows in step. Steering along the use direction (Section 2.7) has little effect in the early layers, with an intact rate of 0.06 at L18, rises through L19 and L20, and produces a full reversal from L21 to L22.

The same band appears on a second surface form. On the scaffold form the class means separate along the use direction at L19 and a linear classifier on the residual reaches an AUC of 0.9 at L21, matching the α/β decay corpus, which indicates that the decision zone is not specific to the surface form.

These layers agree with the main results. The use direction is causal earliest, with a margin reversal already at L20 (Section 3.1.2), its alignment with the output forms over the same layers (Section 3.5), and the norm of the writing direction and the feature span grow into the same band (Sections 3.4 and 3.5).

A sharp onset over a narrow band also appears in a separate task. In the entropy gate the clustering of the residual stream rises at L27 while the causal effect is spread over L26 to L34, so a sharp onset can coexist with a causal effect realised across several layers.

The measurements of Table 11 are correlated, because they are computed from the same activations, so their agreement at L21 marks one band of depth rather than several separate confirmations.

Some onsets are weak, for example the attribution peak at 1.10 times its null, and the causal onset moves with the steering strength. The table supports the claim that commitment forms within a narrow band centred on L21, not that any single measurement is decisive on its own.

C Spinning of directions

The three directions are not fixed across depth. Each is recomputed at every layer, and each rotates as depth increases. This appendix measures how fast each direction rotates and shows that the three rotate at different rates.

We measure the self-alignment of a direction across a depth displacement,

$$\rho(\Delta k) = \left\langle \, |\cos(v^{(\ell)}, v^{(\ell+\Delta k)})| \, \right\rangle_{\ell}, \tag{25}$$

the average over layers of the absolute cosine between the direction at layer ℓ and the same direction Δk layers later. It starts at 1 for $\Delta k = 0$ and decays as the direction reorganises, so a faster decay means faster rotation. One could summarise each curve by the displacement at which ρ falls to one half, but this single number is sensitive to interpolation at the threshold, so we report the full curve $\rho(\Delta k)$ instead.

On the α/β task the three decay curves separate cleanly (Figure 7). The reading direction decays fastest, from $\rho = 0.60$ at one layer to 0.24 at four. The use direction decays slowest, from 0.81 to 0.47. The writing direction lies between, from 0.74 to 0.39. The reading direction therefore rotates fastest and the use direction slowest.

This ordering holds on every concept tested (Figure 8). Across the three forms of the α/β decay corpus and the particle pairs, the self-alignment after four layers is lowest for the reading direction (0.27 to 0.34), highest for the use direction (0.41 to 0.60), and intermediate for the writing direction. The reading direction is the most reorganised and the use direction the most stable, on all of them.

The rotation is real and is not an artefact of the labelling. The reading and writing decays stay above a shuffled-label null, in which the class labels are permuted before the directions are recomputed, so the rotation is not produced by the labels. The use direction is compared instead with a structureless random-walk baseline and stays above it as well.

All three directions rotate, so rotation is a shared background of the task subspace rather than a property of one direction (Sections 3.4 and 3.5), and they separate by the rate of rotation. The reading direction, which is causally inert (Section 3.1), reorganises fastest, while the use direction, which carries the answer, is the most stable across depth.

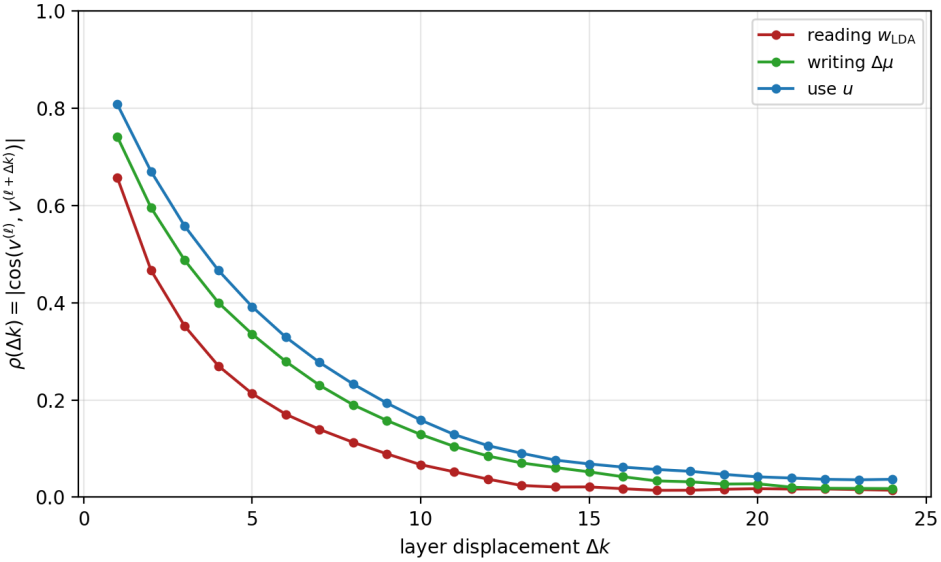


Figure 7: Rotation decay $\rho(\Delta k)$ of the three directions on the α/β task.

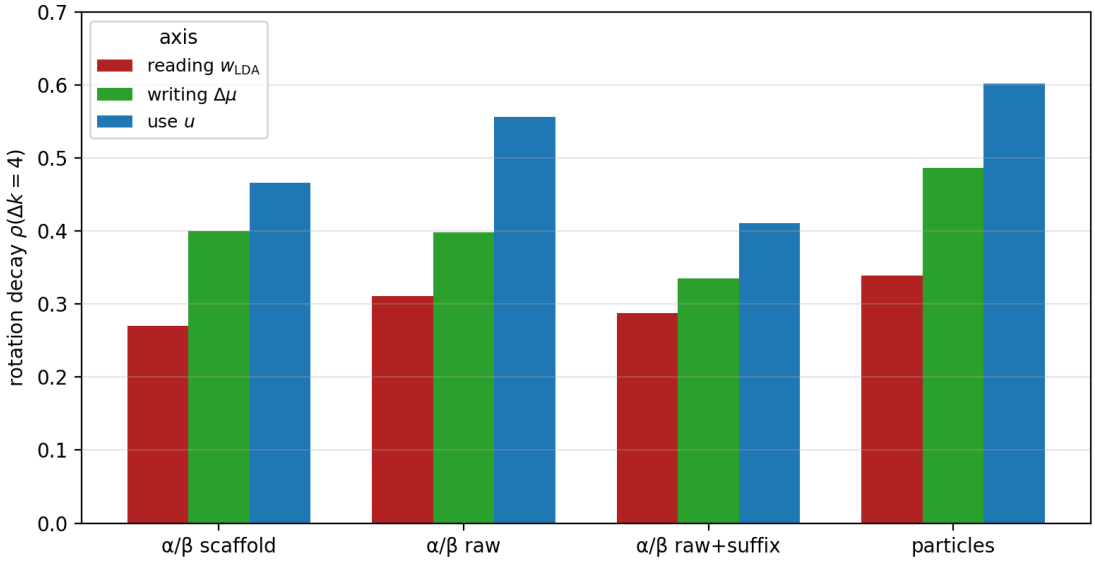


Figure 8: Self-alignment after four layers, $\rho(\Delta k{=}4)$, by concept and direction. The reading direction is lowest and the use direction highest on every concept.

D Geometry of clusters

This appendix describes the procedure behind the interpretation of clusters through physical features (Section 3.6), the construction of the structured cue features, the formation of the clusters, and the criteria for accepting them.

D.1 Prompt cue features

Each prompt of the α/β task is annotated with a set of structured physical cue features that reflect the properties of the described decay, the charge of the emitted particle (for example -1 for β^-), the change in the mass number of the daughter nucleus (for example -4 for α), the change in the charge number (for example $+1$ for β^- and -2 for α), the composition of the emitted particle (for example two protons and two neutrons for α), the presence of an antineutrino, the type of quark transition, and so on. These cue features are grouped into physical families, charge, mass, composite, lepton, and quark. The cue features serve two purposes. They provide a structured annotation for measuring cluster enrichment (Equation 20), and they separate enrichment by a physical feature from enrichment by a surface wording. Because the cue features are physically defined and do not depend on the specific wording of the prompt, the enrichment of a cluster by a cue feature (Table 10) indicates a physical feature rather than a lexical one.

A note on the confound between class and form. The surface families of wording in the α/β corpus are largely single-class (Section 2.3), so an individual cue feature correlates with the class. The enrichment of a cluster by a cue feature, for example charge -1, is therefore statistically related to the class, β. Measuring the cluster activations separately by class (Equation 21) shows, however, that the cluster features activate comparably on both classes (Section 3.6), which separates the detection of a physical feature from the detection of a class. Although the cue feature correlates with the class in the corpus, the cluster activation on it is not confined to the class label.

D.2 Selecting the number of clusters

The clusters used in Section 3.6 are the output of a clustering method with two choices, the algorithm and the number of clusters K. We fix both by a single requirement, that the clusters be stable under paraphrasing of the input, so that a cluster reflects a concept and not the surface wording of the prompts. We test this stability in four steps.

1. **Two paraphrase halves.** Each semantic group in the corpus appears in several wordings. We split the corpus into two halves, one holding the odd wordings and one the even, rebuild the co-importance graph on each half, and re-cluster it. Let $\mathcal{C}$ be the clustering on the full corpus and $\mathcal{C}_A$, $\mathcal{C}_B$ the clusterings rebuilt on the two halves.

2. **Recovery.** For a full cluster $G \in \mathcal{C}$ and a half clustering $\mathcal{C}'$, the recovery is the largest Jaccard overlap with any of its clusters,

$$J(G, G') = \frac{|G \cap G'|}{|G \cup G'|}, \qquad r(G) = \max_{G' \in \mathcal{C}'} J(G, G'),$$

   and the mean recovery is $\bar{r} = \frac{1}{|\mathcal{C}|} \sum_{G \in \mathcal{C}} r(G)$. A high $\bar{r}$ means the clusters survive when the wording changes.

3. **The data-reduction baseline.** A division into halves reduces the recovery on its own, since each half contains fewer prompts, independently of the wording. To isolate this reduction we also partition the corpus at random, ignoring the paraphrase structure, with the same half

sizes. The agreement between two partitions is measured by the adjusted Rand index. For the contingency counts n_{ij} of the features shared by cluster i of one partition and cluster j of the other, with row sums $a_i = \sum_j n_{ij}$, column sums $b_j = \sum_i n_{ij}$, and n features in total,

$$\mathrm{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - \left[\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}\right] / \binom{n}{2}},$$

which equals 1 for identical partitions and is near 0 for independent ones. Over $S = 40$ random partitions we record the agreement between the two halves, $\mathrm{ARI}(\mathcal{C}_A^{(s)}, \mathcal{C}_B^{(s)})$ for $s = 1, \ldots, S$, and their distribution defines the data-reduction baseline. A paraphrase agreement at this baseline indicates that paraphrasing introduces no reduction beyond that of the smaller sample, so the clustering is invariant to wording.

4. **The stability percentile.** We locate the paraphrase agreement $\mathrm{ARI}(\mathcal{C}_A, \mathcal{C}_B)$ within the baseline distribution as a percentile, the fraction of the random partitions that it exceeds,

$$\pi = \frac{100}{S} \sum_{s=1}^{S} \mathbf{1}\left[\mathrm{ARI}(\mathcal{C}_A, \mathcal{C}_B) > \mathrm{ARI}(\mathcal{C}_A^{(s)}, \mathcal{C}_B^{(s)})\right].$$

A percentile near 50 places the paraphrase agreement at the baseline, as stable as a random partition, so the clustering is invariant to wording. A percentile near 0 places it below the baseline, so paraphrasing degrades the clusters.

We first compare the two clustering algorithms. Under the Louvain algorithm the paraphrase agreement is at the 2nd percentile of the random baseline, far below it, so its clusters change with the wording. Under average-linkage agglomerative clustering the agreement is at the 47th percentile, level with the baseline, so its clusters are stable. We therefore use agglomerative clustering.

We then apply the same test to the number of clusters, reading the percentile across a sweep of K (Table 12). At coarse resolutions, $K = 10$ and 11, the percentile is high, 87 and 90, so the clusters are stable but few. At $K = 12$ the percentile is 47, at the baseline, the finest resolution that remains as stable as a random partition. From $K = 13$ upward the percentile falls below the baseline, to 37 at $K = 13$ and to 10 at $K = 14$, so the clusters become sensitive to wording. We therefore cut the hierarchy at $K = 12$, the finest number of clusters that remains stable under paraphrasing. We could instead select K by cluster separation. A standard measure is the silhouette, the mean over features of $(b_i - a_i)/\max(a_i, b_i)$, with a_i and b_i the mean cosine distances from a feature to its own cluster and to the nearest other cluster. It rises with K throughout (Table 12) as finer clusters separate more easily, so it does not bound K, and we keep the selection by stability.

K	paraphrase percentile	silhouette
10	87	0.25
11	90	0.39
12	47	0.48
13	37	0.54
14	10	0.60

Table 12: Stability under paraphrasing across the number of clusters K, for agglomerative clustering on the co-importance graph. The percentile is the position of the paraphrase agreement in the random-partition baseline.

D.3 Cluster acceptance criteria

Each cluster is tested against six criteria and accepted as a candidate when it meets all of them. We write d_i for the direction along which feature i writes and $e(f, p)$ for the effect of ablating feature f on prompt p, the change it produces in the margin of the correct answer over the incorrect one.

1. **Directional consistency.** The features write with a consistent sign rather than cancelling, measured by the mean pairwise signed cosine of their directions,

$$\bar{c}_{\text{sgn}} = \text{mean}_{i<j} \cos(d_i, d_j) \geq 0.50.$$

2. **Coherence.** The features share a common axis, whether they add or oppose along it, measured by the mean pairwise absolute cosine,

$$\bar{c}_{\text{abs}} = \text{mean}_{i<j} |\cos(d_i, d_j)| \geq 0.85.$$

   A high absolute cosine with a low signed cosine marks a cluster whose features share an axis but push against each other.

3. **Semantic specificity.** The cluster is enriched for at least two physical cue features, each with a lift (Equation 20) above 1.5 for at least three values of k.

4. **Cross-form reuse.** The cluster is active across several wordings of the same concept rather than one. A composite reuse score of at least 0.40 combines the diversity of the cluster's effect across the complexity levels, the prompt groups, and the cue types with its balance between keyword-free and keyword-present prompts.

5. **Causal impact.** The joint ablation of the cluster changes the answer or the geometry. The cluster passes if the interaction ratio

$$\text{IR}(C, p) = \frac{e_{\text{joint}}(C, p)}{\sum_{f \in C} e(f, p)},$$

   the effect of ablating all its features together over the sum of their individual effects, is at most 0.50, or if the fraction of prompts whose answer flips under joint ablation exceeds 0.05, or if a test against additivity gives $p < 0.001$.

6. **Level coverage.** The cluster is active for prompts at at least three of the four complexity levels.

Of the twelve clusters of the α/β task (HAC $K = 12$), six are accepted candidates that meet all six criteria including the causal one, three are partial candidates, the multilayer hub whose coherence falls below threshold together with two clusters that meet the five local criteria but fail the causal criterion, their joint ablation flipping fewer than five percent of prompts, and three are descriptive only, the single feature clusters. This ensures that the clusters on which the writing account rests (Section 3.6) are coherent, causal, and functional groups rather than artefacts of the partitioning.

To link the latent organisation of features with the trichotomy of directions, we measure how the ablation of a cluster changes the geometry of the writing and use directions. Ablation is performed by direct subtraction from the true MLP output, after which the change of the $\Delta\mu$ and u directions across layers is recorded against a null distribution built from random groups of the same size and from the same layers.

D.4 Cluster ablation

The change of a direction under ablation is measured by the norm-free angle

$$\theta^{(\ell)} = \arccos\left(\langle \hat{v}^{(\ell)}_{\text{before}}, \hat{v}^{(\ell)}_{\text{after}} \rangle\right),$$

the angle between the unit direction before and after ablation.

Ablation of clusters rotates the writing direction $\Delta\mu$. The change in norm is small and always corresponds to compression rather than growth. On the layers where the ablation has an effect, that effect is almost entirely a rotation, about 99 percent of it, and not a change in norm.

The strongest rotations are produced by the early clusters. The charge cluster, at L10 with the rotation measured across L10 to L13, rotates the writing direction by about 14.6 degrees and the lepton number cluster by about 12.7 degrees. Intermediate rotations come from the late output cluster, about 9.2 degrees, and the cluster at L24, about 8.2 degrees. Single clusters at the intermediate layers produce small rotations of about 1 to 3 degrees. This pattern, in which the early clusters rotate the writing direction most, is consistent with the writing being formed early (Section 3.2) and used causally later (Section 3.1).

At most layers the ablation of a single feature changes the writing direction by about 1 degree, in the range 0.8 to 1.7 degrees, so the writing direction is largely distributed and is not localised to any single feature. The null for single features is not reliably estimated because the pool of single feature groups is small.

D.5 Writing direction rotation

In the α/β decay corpus the writing and use directions are close to orthogonal, with $\cos(\Delta\mu, u)$ about 0.008, unlike on the scaffold form (Section 3.4) where they are partly aligned. Cluster ablation rotates the writing direction by up to about 15 degrees (Section D.4), but this rotation is not directed toward the use direction, and therefore not toward the output, since the use direction converges to the unembedding (Section 3.1). The change it produces in the alignment of the two directions, $\mathrm{d}\cos(\Delta\mu, u)$, lies between about -0.009 and $+0.019$ across the clusters and does not exceed the null covariance, whose 95th percentile is about 0.037 to 0.039. The strongest change toward use, from the early charge cluster at about $+0.018$, is still below this threshold, and the remaining clusters give changes of order ± 0.001 to 0.006 with no stable sign. The near orthogonality is therefore preserved, the rotation takes place in components other than the use direction, and the separation of writing and use is not held up by any single cluster whose removal would destroy it.

This is consistent with the convergence of the writing direction to the output (Section 3.1) rather than in tension with it. That convergence is a depth-distributed process not attributed to individual clusters, matching the distributed picture of Section D.4, in which the writing direction is not carried by any single feature.

E Whitening turns the optimal probe from the use direction

Section 3.1.3 established that the optimal linear decoder w_{LDA} is almost orthogonal to the use direction u. Here we show that this orthogonality follows from the definition of the probe. The probe is $w \propto \Sigma^{-1}\Delta\mu$ of Equation 7, where $\Delta\mu$ is the difference of class means and Σ is the within-class covariance with the diagonal regularisation of Section 2.5.2. All quantities are computed on the train split with the same regularisation to which the probe is fitted. For a unit direction $\widehat{u}$ we write $\sigma_u^2 = \widehat{u}^\top \Sigma \widehat{u}$ for the variance of the representation along $\widehat{u}$.

Let $\widehat{u}$ first be an eigenvector of Σ with eigenvalue σ_u^2. Using $\langle \Sigma^{-1}\Delta\mu, \widehat{u}\rangle = \langle \Delta\mu, \Sigma^{-1}\widehat{u}\rangle = \sigma_u^{-2}\langle \Delta\mu, \widehat{u}\rangle$ and dividing by $\|\Sigma^{-1}\Delta\mu\| = \|\Delta\mu\|/\lambda_{\mathrm{eff}}$,

$$\cos(w, \widehat{u}) = \cos(\Delta\mu, \widehat{u})\, \frac{\lambda_{\mathrm{eff}}}{\sigma_u^2}, \qquad \lambda_{\mathrm{eff}} = \frac{\|\Delta\mu\|}{\|\Sigma^{-1}\Delta\mu\|}. \tag{26}$$

The alignment of the probe with an axis equals the alignment of the raw difference of means divided by the variance along that axis. For a general unit $\widehat{u}$ that is not an eigenvector, set $r = \Sigma\widehat{u} - \sigma_u^2\widehat{u}$. Then $\Sigma^{-1}\widehat{u} = (\widehat{u} - \Sigma^{-1}r)/\sigma_u^2$, and substitution into the second step gives the exact equality

$$\cos(w, \widehat{u}) = \cos(\Delta\mu, \widehat{u})\, \frac{\lambda_{\mathrm{eff}}}{\sigma_u^2} - \frac{\langle \widehat{w}, r\rangle}{\sigma_u^2}. \tag{27}$$

The first term is a prediction and the second a computable residual. The scale λ_{eff} is small, since in the eigenbasis of Σ one has $\|\Sigma^{-1}\Delta\mu\|^2 = \sum_i d_i^2/\lambda_i^2$ with d_i the coordinates of $\Delta\mu$, so λ_{eff} is a weighted power mean of order minus two of the eigenvalues and is dominated by the smallest of them. After whitening the probe concentrates in the low-variance subspace, and the ratio $\lambda_{\mathrm{eff}}/\sigma_u^2$ is small for any axis with appreciable variance.

The use direction is such an axis. Its variance sits at the 98.4th and 98.7th percentile of the spectrum of Σ on the two concepts, the factor $\lambda_{\mathrm{eff}}/\sigma_u^2$ in the decision zone is 0.059 on the decay task and 0.031 on number agreement, and the prediction of Equation 26 agrees with the measured alignment to within the residual of Equation 27. The per-layer prediction error is 0.039 and 0.044, and the measured residual $\langle \widehat{w}, r\rangle/\sigma_u^2$ is 0.038 and 0.048, so the prediction error is the residual term of the exact equality (Table 13).

concept	$\cos(\Delta\mu, u)$	percentile of σ_u^2	$\lambda_{\mathrm{eff}}/\sigma_u^2$	prediction error	residual
decay α/β	0.28	98.4	0.059	0.039	0.038
number agreement	0.18	98.7	0.031	0.044	0.048

Table 13: The attenuation identity in the decision zone, medians on the train split. The alignment $\cos(\Delta\mu, u)$ before whitening, the percentile of the variance σ_u^2 along u in the spectrum of Σ, the attenuation factor of Equation 26, the per-layer error of that prediction, and the exact residual of Equation 27. The prediction error equals the residual, so the identity holds to the residual term.

The variance along u is large for the following reason. The margin is a nonlinear function of the layer state, $\mathrm{margin}(p) = M(h(p))$. A first order expansion about the class mean gives $\mathrm{margin}(p) \approx M(\mu_c) + \langle \nabla M(\mu_c), h(p) - \mu_c\rangle$, and if the gradient does not vary from prompt to prompt, $g(p) \approx \bar{g}$, the within-class fluctuation of the margin is the projection of the state onto $\bar{g}$. The variance of that projection is a quadratic form of the covariance,

$$\mathrm{Var}_c(\mathrm{margin}) = \bar{g}^\top \Sigma\, \bar{g} = \|\bar{g}\|^2 \sigma_u^2, \qquad \widehat{u} = \bar{g}/\|\bar{g}\|. \tag{28}$$

The within-class fluctuations of the confidence of the model lie along the direction by which it reads the answer, and that direction therefore carries a large within-class variance. Both approximations

are checked by the ratio $\mathrm{Var}_c(\mathrm{margin})/(\|\bar{g}\|^2\sigma_u^2)$, which is one where they hold. At the readout layer it is 1.06 and 1.16 and Equation 28 is exact, while in the decision zone it rises to 20.1 and 3.9, since the gradient varies from prompt to prompt and adds variance to the margin beyond the one accounted for, so in the zone Equation 28 gives a lower component of the margin variance and the high variance of u rests on the measured percentile.

The result is as follows. The exact equality of Equations 26 and 27 turns the optimal probe from any high-variance axis by a computable factor, the measurement shows that the use direction is such an axis, and Equation 28 explains why. The use of a direction is accompanied by variance along it, which hides it from the optimal linear reader.

A remark on the regularisation. The weight $s = 0.1$ is fixed a priori and shared by all analyses. As it is raised the attenuation weakens toward the ratio of the diagonal scales, so the effect is not created by the regularisation, and the value reported at $s = 0.1$ is a conservative one that an unregularised probe would only strengthen.

F Reproduction of circuit tracing

F.1 Staging and pipeline

The first phase of the project reproduced the attribution-graph methodology of [20] and the analytical style of [21] on an open model: Qwen3-4B (base) with per-layer transcoders [29] in place of the cross-layer transcoders of the original. The pipeline was: baseline runs and fixing of the answer position; extraction of active transcoder features; construction of an attribution graph, with edges given by gradient times activation; validation of nodes and edges by targeted interventions (feature ablation and patching); assembly of family chains and comparison across surface framings. The intervention protocol and the fixing of the answer position follow the canonical setup used throughout the main text.

F.2 What the reproduction revealed

Four observations of this phase determined the programme of the main part.

(i) Distribution with low necessity. The necessity of individual paths is systematically low (10 to 28 percent across corpora) while the total causality of the chain is high: the causal structure is a property of the ensemble, not of the listed units. This observation became the distributed, non-localised picture of Section 3.6 and Appendix D, and it fixed the choice of the cluster, rather than the feature, as the minimal causal unit.

(ii) The attribution label is not the causal direction. A sign-complete ablation, over all features rather than the positive ones only, revealed a systematic divergence: the β-prompt sign-flip rate at layer 24 is 25.4 percent against 0.4 percent on α, which the attribution labels do not predict. The attribution labels of the graph proved an unreliable indicator of causal role, which motivated setting them aside in the main part (Section 3.6) in favour of directly measured geometric objects: the reading, writing, and use directions.

(iii) A chain can be present while its output is suppressed. Cross-level convergence of the circuits (interchange similarity at least 0.74, including levels of zero accuracy) and carrier features with near-constant activation yet non-zero causal effect ($\Delta\text{act} \approx 0$, $\Delta\text{eff} = +1.16$) showed that the presence and readability of a structure do not predict its contribution to the answer. This is an early form of the central dissociation between what is decoded and what is used (Sections 3.1.3 and 3.5).

Taken together, the reproduction succeeded as a method and proved insufficient as an explanation: the attribution graphs answered through which nodes the computation flows, but not in what geometry the decision is read, written, and used. The main part answers this second question.